

1. Уровень клетки

Начальная система уравнений

$$\frac{dG}{dt} = \theta - k_{pH}GC - \alpha G$$

$$\frac{dG_D}{dt} = mP\theta - k_{pD}G_D C - \alpha G_D$$

$$\frac{dC}{dt} = \eta\theta - (k_{pH}G + k_{pD}G_D)C - \beta C$$

Ищем стационарное состояние:

$$\frac{d...}{dt} = 0:$$

$$G = \frac{\theta}{\alpha + k_{pH}C}$$

$$G_D = \frac{mP\theta}{\alpha + k_{pD}C}$$

$$C = \frac{\eta\theta}{k_{pH}G + k_{pD}G_D + \beta}$$

Эта система преобразуется в уравнение:

$$y^3 \gamma \kappa + y^2 (\gamma(1 + mP - \eta) + \kappa(1 + \gamma)) + y(1 + mP\gamma + \kappa - \eta(1 + \gamma)) - \eta = 0$$

$$\text{Где } y = \frac{Ck_{pH}}{\alpha}, \quad \gamma = \frac{k_{pD}}{k_{pH}}, \quad \kappa = \frac{\beta\alpha}{\theta k_{pH}}$$

При $\kappa \ll 1$, $\eta < 1 + mP$ и $\gamma > 0$:

$$y = \frac{-(1+mP-\eta-\eta\gamma) + \sqrt{(1+mP-\eta-\eta\gamma)^2 + 4\eta\gamma(1+mP-\eta)}}{2\gamma(1+mP-\eta)}$$

При $\eta \geq 1 + mP$ или $\gamma < 0$: $y \rightarrow \infty$

(Графики в приложении - номер 1..)

Burst-size:

$$n = \frac{k_{pH}[GC]_{P=0}}{\delta}$$

$$\psi_m n = \frac{k_{pH}GC}{\delta}$$

$$\rho_m \psi_m n = \frac{k_{pD}G_D C}{\delta}$$

Burst-size через параметры клетки:

$$n = \frac{\theta}{\delta} \frac{y}{1+y} \text{ при } P = 0$$

$$\psi_m n = \frac{\theta}{\delta} \frac{y}{1+y}$$

$$\rho_m \psi_m n = \frac{mP\theta}{\delta} \frac{\gamma y}{1+\gamma y}$$

(Графики в приложении - номер 2..)

2. Уровень организма

$$\frac{dT}{dt} = b - (d(1 - h) + kV + kW)T$$

$$\frac{dI}{dt} = kVT - \delta I$$

$$\frac{dP_m}{dt} = kW P_{m-1} - (d(1 - h) + kV + kW)P_m$$

$$\frac{dD_m}{dt} = kV P_m - \delta D_m$$

$$\frac{dV}{dt} = n\delta I + n\delta \sum_{m=1}^{\infty} \psi_m D_m - cV$$

$$\frac{dW}{dt} = n\delta \sum_{m=1}^{\infty} \rho_m \psi_m D_m - cW$$

$$v = \frac{kV}{d}$$

$$w = \frac{kW}{d}$$

$$q = \frac{w}{1-h+v+w}$$

$$R_0 = \frac{nk b}{c d}$$

Ищем стационарное состояние:

$$\frac{d...}{dt} = 0:$$

$$T = \frac{b}{d(1-h+v+W)}$$

$$I = \frac{kVT}{\delta}$$

$$P_m = \frac{v}{1-h+v+w} P_{m-1} = q^m T$$

$$D_m = \frac{kVP_m}{\delta} = q^m I$$

$$1 - h + v + w = R_0(1 + \sum_{m=1}^{\infty} \psi_m q^m)$$

$$(1 - h + v + w)^2 = R_0 \frac{v}{q} \sum_{m=1}^{\infty} \rho_m \psi_m q^m$$

система уравнений:

$$1 - h + v + w = R_0(1 + \sum_{m=1}^{\infty} \psi_m q^m)$$

$$(1 - h + v + w)^2 = R_0 \frac{v}{q} \sum_{m=1}^{\infty} \rho_m \psi_m q^m$$

$$q = \frac{w}{1-h+v+w}$$

Уравнения для решения в программе:

$$(R_0(1 + \sum_{m=1}^{\infty} \psi_m q^m))^2 = R_0 \frac{R_0(1 + \sum_{m=1}^{\infty} \psi_m q^m)(1-q)+h-1}{q} \sum_{m=1}^{\infty} \rho_m \psi_m q^m$$

$$A = R_0(1 + \sum_{m=1}^{\infty} \psi_m q^m)$$

$$w = qA$$

$$v = A(1 - q) + h - 1$$

(графики в приложении - номер 3..)

3.S_eff

$$n \rightarrow n(1 + \Delta_n)$$

$$\psi_m \rightarrow \psi_m(1 + \Delta_\psi)$$

$$\rho_m \rightarrow \rho_m(1 + \Delta_\rho)$$

$$I^{mut}(t) = I^{ss}x(t)$$

$$D_m^{mt}(t) = D_m^{mut}y_m(t)$$

$$V^{mut}(t) = V^{ss}z(t)$$

$$\frac{dx}{dt} = (z - x)\delta$$

$$\frac{dy_m}{dt} = (z - y_m)\delta$$

$$z + \frac{1}{c} \frac{dz}{dt} = \frac{R_0(1+\Delta_n)}{1-h+v+w} (x + \sum_{m=1}^{\infty} \psi_m (1 + \Delta_{\psi}) q^m y_m)$$

$$z = (1 + s_{eff})x = (1 + s_{eff})y_m$$

$$s_{eff} = \Delta_n + \frac{\sum_{i=1}^{\infty} \psi_i q^i \Delta_{\psi i}}{1 + \sum_{i=1}^{\infty} \psi_i q^i}$$

Δ_n и $\Delta_{\psi n}$: для κ

$$\Delta_n = \frac{\partial n}{\partial \kappa} = \frac{\partial \ln n}{\partial \kappa} \partial \kappa = \partial \kappa \frac{\partial}{\partial \kappa} \ln\left(\frac{y}{1+y}\right)_{P=0}$$

$$\Delta_{\psi n} + \Delta_n = \partial \kappa \frac{\partial}{\partial \kappa} \ln\left(\frac{y}{1+y}\right)$$

$$\Delta_{\psi n} = \partial \kappa \frac{\partial}{\partial \kappa} \ln\left(\frac{y}{1+y}\right) - \partial \kappa \frac{\partial}{\partial \kappa} \ln\left(\frac{y}{1+y}\right)_{P=0}$$

Δ_n и $\Delta_{\psi n}$: для η

$$\Delta_n = \frac{\partial n}{\partial \eta} = \frac{\partial \ln n}{\partial \eta} \partial \eta = \partial \eta \frac{\partial}{\partial \eta} \ln\left(\frac{y}{1+y}\right)_{P=0}$$

$$\Delta_{\psi n} + \Delta_n = \partial \eta \frac{\partial}{\partial \eta} \ln\left(\frac{y}{1+y}\right)$$

$$\Delta_{\psi n} = \partial \eta \frac{\partial}{\partial \eta} \ln\left(\frac{y}{1+y}\right) - \partial \eta \frac{\partial}{\partial \eta} \ln\left(\frac{y}{1+y}\right)_{P=0}$$

Δ_n и $\Delta_{\psi n}$: для γ

$$\Delta_n = \frac{\partial n}{\partial \gamma} = \frac{\partial \ln n}{\partial \gamma} \partial \gamma = 0$$

$$\Delta_{\psi n} + \Delta_n = \partial \gamma \frac{\partial}{\partial \gamma} \ln\left(\frac{y}{1+y}\right)$$

$$\Delta_{\psi n} = \partial \gamma \frac{\partial}{\partial \gamma} \ln\left(\frac{y}{1+y}\right)$$

(графики в приложении - номер 4..)